Precision of the $V \to q\bar{q}g$ evaluator on the LHC Z+jet phase space

Variables. In a production region $A > 1$ ($A = x, y, z$; the other two invariants are negative) we use

$$v = \frac{1}{A} = \frac{q^2}{\hat{s}}, \qquad u = -\frac{b}{A}, \qquad w = -\frac{c}{A} = 1 - u - v, \qquad u, v, w > 0,$$

for region z: $u = -y/z$, $w = -x/z$ (the package's region triangle is $(X, Y) = (w, u)$, $Z = v$). With E_j, θ the jet energy and angle in the partonic frame, $u = E_j(1 + \cos\theta)/\sqrt{\hat{s}}$ and $w = E_j(1 - \cos\theta)/\sqrt{\hat{s}}$, so

$$\frac{p_T^2}{M_Z^2} = \frac{u\,w}{v}, \qquad \frac{u}{w} = e^{2\eta^*}, \qquad \sqrt{\hat{s}} = \frac{M_Z}{\sqrt{v}} \qquad (q^2 = M_Z^2),$$

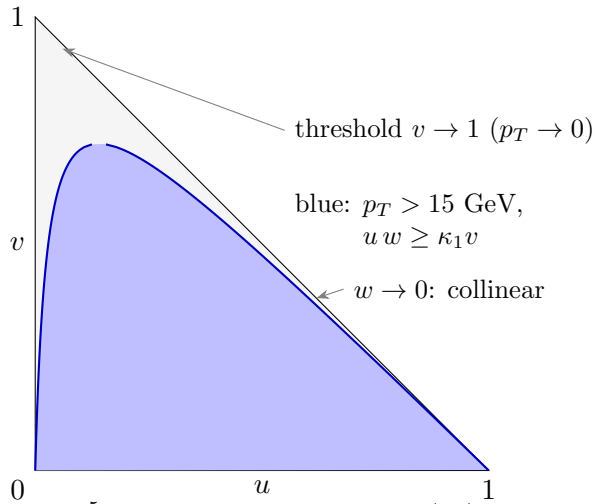
with η^* the jet rapidity in the partonic frame. Every edge of the triangle is a physical limit:

- **Threshold**, $v \to 1$: then $u + w \to 0$, so $p_T \to 0$.
- **Collinear**, $u \to 0$ or $w \to 0$: $p_T \to 0$, or a forward jet ($|\eta^*| \to \infty$).
- **High energy**, $v \to 0$.

The LHC domain. We take $15\,\text{GeV} < p_T < 2\,\text{TeV}$, $\sqrt{\hat{s}} \le 13.6$ TeV (the collider energy) and the jet acceptance $|\eta^*| \le 4.5$. In (u, v), with $\kappa_1 = (15/91.19)^2 = 0.0271$ and $\kappa_2 = (2000/91.19)^2 = 481$:

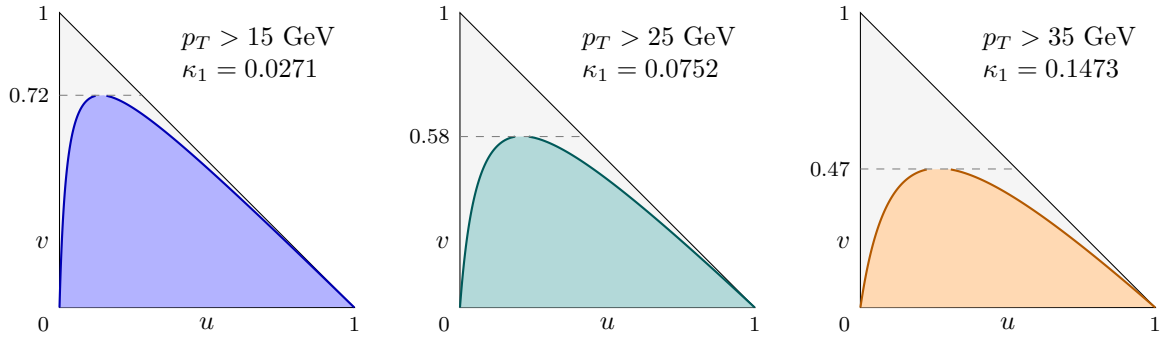
$$\kappa_1\, v \;\le\; u\, w \;\le\; \kappa_2\, v, \qquad v \ge \left(\frac{M_Z}{13.6\,\text{TeV}}\right)^2 = 4.5 \cdot 10^{-5}, \qquad e^{-9} \le \frac{u}{w} \le e^{9}.$$

The p_T cut keeps away from threshold, $v \le 1 + 2\kappa_1 - 2\sqrt{\kappa_1 + \kappa_1^2} = 0.72$. The rapidity cut keeps away from the collinear edges, $u, w \ge 1.2 \cdot 10^{-4}$. Without it, the p_T cut alone allows u, w down to $\kappa_1 v_{\min} \approx 1.2 \cdot 10^{-6}$: a 15 GeV jet at the highest energies, with $\eta^* \approx 7$. The rapidity cut is applied in the partonic frame. In the high-energy corner, where it matters, the boost of the partonic system is small ($\le \ln(\sqrt{s}/\sqrt{\hat{s}})$), so η^* is close to the lab rapidity.



high energy $v \to 0$: $v \ge 4.5 \cdot 10^{-5}$; collinear edges at $v \to 0$: $|\eta^*| \le 4.5$ keeps $u, w \ge 1.2 \cdot 10^{-4}$

The energy and rapidity cuts act at the 10^{-4} scale, too close to the corners to be seen on this linear scale.



The domain $u\,w \geq \kappa_1 v$ for $p_{T,\min} = 15$, 25 and 35 GeV, with $\kappa_1 = (p_{T,\min}/M_Z)^2$. The largest v (dashed), i.e. the closest approach to threshold, is 0.72, 0.58 and 0.47. At high energy every cut still reaches the collinear edges, as $u, w \approx \kappa_1 v$; there the rapidity acceptance takes over.

How the accuracy is estimated. Two different quantities enter.

1. The error estimate of the double evaluator (returned by the package with every value):

$$\varepsilon_{\text{est}} = 10\,|R_1 - R_2| + \epsilon \sum |\text{terms}|, \qquad \epsilon = 10^{-13}.$$

- R_1, R_2 are two double evaluations of the same point, from the same 27-digit start values, along two different complex-deformed integration paths. Their difference catches the path-dependent integration error.
- $\sum |\text{terms}|$ is the sum of the absolute values of all terms (rational function × generator × multiplier) in the remainder. The second term covers the systematic error of the double system, which is the same on every path (its matrices come from a floating-point orthonormalisation), amplified by the cancellation in the sum. ϵ is provisional: the smallest value covering every value checked so far.

In all tests this was an upper bound, but about 1000 times too large. Here it only selects the points to check.

2. The actual error is the difference from a quad evaluation at the same point: $|R_{\text{double}} - R_{\text{quad}}|/|R_{\text{quad}}|$. Quad integrates the exact integer system in double-double from the 27-digit start values, and builds the generators, the rational functions and the sums in double-double. It is trusted to far beyond the 10^{-6} level:

- its own two-path error is $\sim 10^{-23}$;
- it agrees with the exact published planar values to $3 \cdot 10^{-16}$, the precision of that reference;
- in the production regions it agrees with a direct integration from the decay to 10^{-23}.

3. The check (`accuracy_study/pheno_domain_check.py`):

- **Points:** about 330 per region, log-spaced in v and dense towards both collinear edges.
- **Double:** every point, with its estimate. All 84 values are included: 6 helicity coefficients ($\alpha_{1,2,3}$, $\gamma_{1,2,3}$) times 14 colour layers.

- **Quad:** where the estimate exceeds 10^{-6}, the three worst points (by estimate) of each band of $\min(u, w)$ in each region. That is 9 points per region with the rapidity cut and 15 without.
- **Colour sum:** for the colour-summed coefficients ($N_c = 3$, $N_f = 5$, $N_{f,Z} = 1$, the worst of the six) we quote the bound $\sum_l |c_l|\,|\delta_l| / |\sum_l c_l\, a_l|$. Here $c_l = N_c^a N_f^b N_{f,Z}^c$ and δ_l is the actual error of layer l; by the triangle inequality this is an upper bound.

Limits.

- Quad was run only at a selection of points, the worst by estimate. The estimate ranks the points well, but the real worst point could be one not checked.
- $N_{f,Z} = 1$ is an arbitrary choice for the colour sum.
- The colour-sum figure is a bound and can overstate the error.
- To make the statement airtight, run quad at every point whose estimate exceeds 10^{-6}: 250–300 points per region, 1–2 hours in total.

Results. Error of the colour-summed coefficients in double at the checked points (the worst of each band, in all three regions):

$\min(u, w)$	without rapidity cut	with $\lvert\eta^*\rvert \le 4.5$
$\geq 10^{-3}$	10^{-8} – $6 \cdot 10^{-6}$	median $1.2 \cdot 10^{-6}$, worst $3.4 \cdot 10^{-6}$
10^{-4} – 10^{-3}	$4 \cdot 10^{-7}$ – $9 \cdot 10^{-6}$	median $1.1 \cdot 10^{-6}$, worst $8.7 \cdot 10^{-6}$
$2 \cdot 10^{-5}$ – 10^{-4}	$3 \cdot 10^{-5}$ – 10^{-4}	outside the domain
$< 2 \cdot 10^{-5}$	10^{-3} – $O(1)$	outside the domain

- **The failing corner.** At high energy $\min(u, w) \approx p_T^2/\hat{s}$, so the points that fail ($\min(u, w) \lesssim 10^{-4}$) are jets with $\sqrt{\hat{s}} \gtrsim 100\, p_T$, i.e. $\eta^* \gtrsim 4.6$. They are outside the detector acceptance, and the rapidity cut removes them.
- **Individual colour layers.** Some single layers show relative errors up to 10^{-3} even at high p_T. They are 10^4–10^6 times smaller than the other layers at the same point: relative to the largest value there, the error is at most $4 \cdot 10^{-6}$ (typically 10^{-7}), and they do not affect the colour sum.
- **Where double loses digits.** The sum over rational functions times generators cancels: the sum of |terms| exceeds the value by 10^4 in the bulk and up to 10^9 near the edges. Where double is not enough, quad (`precision='quad'`, 5–10 s per point) keeps 15 or more digits. Near the edges, the corner series is the proper tool.

Conclusion. On the LHC domain ($15\,\text{GeV} < p_T < 2$ TeV, $\sqrt{\hat{s}} \le 13.6$ TeV, $|\eta^*| \le 4.5$) the double evaluator gives the colour-summed coefficients to better than 10^{-5} in all three regions: typically 10^{-6}, at worst $9 \cdot 10^{-6}$, at the forward edge of the acceptance ($p_T = 15$–55 GeV, $\min(u, w) \approx 10^{-4}$) and at the highest p_T and energy ($p_T \approx 1.4$ TeV, $\sqrt{\hat{s}} = 13.6$ TeV). Relative to the largest of the 84 values at a point, the error is at most $4 \cdot 10^{-6}$. A strict 10^{-6} everywhere needs quad at those points.